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CHAPTER 1

INTRODUCTION

1.1 Overview of LoRa Technology

Reliable communication between vehicles can be difficult in remote areas where cellular networks, Wi-Fi, and wired connections are unavailable. LoRa provides a low-power way to exchange small amounts of data over long distances, making it useful for such environments.

LoRa performance depends on factors such as terrain, antenna design, and radio settings. One important setting is the spreading factor (SF), which affects data rate and transmission time. A higher spreading factor generally improves reception of weak signals but increases transmission time, while a lower spreading factor allows faster communication.

In this project, ESP32 microcontrollers are connected to Reyax RYLR998 LoRa modules, which are configured using UART AT commands. The firmware is developed using ESP-IDF. The system consists of four nodes: one master and three additional nodes. The master discovers reachable nodes by broadcasting messages and collecting acknowledgements. When some nodes do not respond, selected reachable nodes act as submasters and assist in searching for them, helping extend the discovery process.

1.2 Problem Statement

In infrastructure-constrained environments, mobile vehicles may not be able to communicate directly with every node in a wireless network because of limited radio coverage, obstacles, or changing link conditions. A network that relies only on direct communication with a master node may therefore fail to discover or reach all participating nodes. The problem is to develop a master-coordinated dynamic routing mechanism using LoRa communication that identifies unresponsive nodes, delegates discovery tasks to reachable nodes, and establishes routes through intermediate nodes where possible. The proposed system is implemented using ESP32 microcontrollers and Reyax RYLR998 LoRa modules, with experimental evaluation of communication distance, bitrate, and transmission time at Spreading Factors 8 and 9

1.3 Motivation

The motivation for this project comes from the need for reliable communication between vehicles in remote areas where cellular networks, Wi-Fi, and other communication infrastructure may not be available. In such situations, a vehicle may be unable to communicate directly with every other node because of distance or obstacles.

LoRa offers a promising solution for long-range, low-power communication. However, establishing radio communication alone is not enough when some nodes cannot be reached directly. This project explores a cooperative approach in which reachable nodes assist the master in discovering other nodes, with the aim of improving network connectivity and supporting dynamic route recovery.

The project also provides an opportunity to study the practical performance of LoRa communication. By measuring communication distance, bitrate, and transmission time for Spreading Factors 8 and 9, the system helps examine how radio configuration affects communication performance. Through this work, the project combines embedded-system development, wireless communication, and routing concepts to investigate a flexible communication network for infrastructure-constrained environments.

1.4 Objectives of the Project

The main objective of this project is to design and implement a four-node, master-coordinated LoRa communication prototype using ESP32 microcontrollers and Reyax RYLR998 modules for communication in environments where conventional infrastructure may be unavailable. The specific objectives are:

1. Development of a four-node communication system: Interface ESP32 microcontrollers with RYLR998 LoRa modules and establish wireless communication through UART-based module configuration and control.

2. Implementation of node discovery: Develop a master-based discovery procedure in which the master sends discovery messages and identifies reachable nodes through their acknowledgement responses.

3. Development of cooperative node searching: Enable selected responding nodes to act as submasters and assist the master in searching for nodes that have not responded directly.

4. Implementation of packet-handling mechanisms: Develop procedures for message addressing, packet parsing, acknowledgements, response timeouts, retries, and randomized backoff, as required by the protocol.

5. Performance comparison of Spreading Factors 8 and 9: Measure communication distance, bitrate, and transmission time under the selected experimental conditions.

6. Analysis of experimental results: Compare the collected measurements to understand the effect of spreading-factor selection on communication performance.

7. Testing of the proposed system: Evaluate the four-node prototype by observing packet exchanges, node responses, and the operation of the cooperative discovery mechanism. .

1.5 Scope of the Project

This project focuses on developing a four-node LoRa communication prototype using ESP32 microcontrollers and Reyax RYLR998 LoRa modules. The firmware is developed using ESP-IDF, with UART-based AT commands used to configure and control the modules.

The project implements the developed node-discovery and communication procedures. Experimental testing compares Spreading Factors 8 and 9 by measuring communication distance, bitrate, and transmission time.

The scope is limited to the four-node prototype and the experiments conducted under the selected testing conditions. The results provide a basis for understanding LoRa communication performance and exploring the proposed network approach.

CHAPTER 2

LITERATURE REVIEW

2.1 LoRa Communication and Network Architecture

LoRa (Long Range) is a wireless communication technology designed for long-distance communication with relatively low power consumption. Augustin et al. [1] studied LoRa modulation, its physical-layer characteristics, data rate, receiver sensitivity, and communication range. Their experimental results showed that spreading factor and data rate significantly influence communication performance. This research established an important foundation for understanding the trade-offs involved in selecting LoRa communication parameters.

Traditional LoRaWAN networks generally use a star-of-stars architecture in which end devices communicate with gateways that connect to a network server. Although this arrangement is suitable for many monitoring applications, it relies on gateway infrastructure and does not inherently provide direct multi-hop communication between end devices. For applications requiring communication without a central gateway, an alternative networking protocol can be developed using LoRa radio links.

The Reyax RYLR998 module used in this project supports wireless communication through UART-based AT commands. Its configuration includes device addressing, network identification, frequency selection, and radio parameters [9]. The ESP32 handles the application-level logic, allowing the nodes to exchange discovery messages and acknowledgements according to the developed protocol.

2.2 LoRa Mesh Networking and Multi-Hop Communication

Cotrim and Kleinschmidt [2] reviewed different approaches to multi-hop communication in LoRaWAN networks. They explained how intermediate devices can forward messages to extend network coverage beyond the direct communication range of a gateway. Their review also highlighted important design considerations, including routing, network reliability, latency, and energy consumption.

Lundell et al. [3] proposed a routing protocol for LoRa mesh networks to overcome

the limitations of single-hop communication. Their work included a proof-of-concept implementation that was evaluated through laboratory tests and a field trial. The study demonstrated the feasibility of using a routing protocol to enable multi-hop communication over LoRa links.

Berto et al. [4] presented a LoRa-based peer-to-peer mesh network that operated without relying on LoRaWAN gateways. Their hardware and software prototype investigated multi-hop communication between nodes, demonstrating how intermediate nodes can extend network reachability.

More recently, Almeida et al. [7] developed a gateway-free mesh system using ESP32-based hardware and LoRa communication. Their design included neighbour-based routing, hop-by-hop acknowledgements, and mechanisms for handling communication failures. Their experimental evaluation considered normal conditions, intermediate-node failures, and urban deployments.

These studies show that LoRa can support networking approaches beyond conventional single-hop communication. They also demonstrate that node forwarding, routing decisions, and failure handling need to be addressed at the protocol level. The present project investigates a master-coordinated approach in which reachable nodes assist in discovering nodes that do not respond directly to the master.

2.3 Dynamic Routing and Cooperative Node Discovery

Routing is an important component of a multi-node wireless network because it determines how information can reach its destination when direct communication is unavailable. Existing approaches include proactive routing, which maintains route information in advance; reactive routing, which discovers routes when required; and hybrid routing, which combines aspects of both methods.

Wong et al. [5] reviewed developments in multi-hop and mesh LoRa networks, including routing protocols, energy efficiency, communication access, and security. Their review discussed proactive, reactive, and hybrid approaches and identified route selection and network scalability as important considerations in practical deployments.

Pueyo Centelles et al. [6] proposed a minimalistic distance-vector routing protocol for LoRa mesh networks. Their work investigated the use of time-on-air as a routing metric to account for differences in spreading factor and link characteristics. The authors evaluated their approach using simulations and compared it with other routing strategies. The study

illustrates how routing decisions can consider communication cost rather than relying solely on the number of hops.

Solé et al. [8] introduced LoRaMesher, an open-source library for multi-hop mesh networks on ESP32 microcontrollers. The library uses proactive distance-vector routing and maintains routing tables to support communication between nodes without depending on LoRaWAN gateways. This work provides a practical example of implementing routing functionality on embedded hardware.

In the proposed project, the master first broadcasts a discovery message and records the nodes that respond. If some expected nodes remain undiscovered, selected responding nodes can receive search requests and assist in locating them. These nodes act as submasters during the discovery process. This is a cooperative, master-coordinated approach rather than the fully distributed routing-table approach used by some existing mesh implementations.

The distinction is important: discovering a node through a helper is not, by itself, proof that subsequent data packets can be forwarded through that helper. Multi-hop routing requires an appropriate forwarding mechanism in addition to the discovery procedure.

2.4 Spreading Factor, Bitrate and Communication Range

The spreading factor is a key parameter in LoRa communication because it influences receiver sensitivity, data rate, and packet transmission time. Augustin et al. [1] investigated the relationship between LoRa radio parameters and communication performance through experimental testing and analysis.

Guo et al. [10] experimentally studied the effect of physical-layer parameters, including spreading factor, bandwidth, coding rate, and packet length, on LoRa communication performance. Their study showed that radio configuration affects communication reliability and transmission time. Higher spreading factors can improve sensitivity under suitable conditions, but they generally require more time to transmit the same payload.

The Reyax RYLR998 AT Command Guide [9] also specifies that increasing the spreading factor improves sensitivity while increasing transmission time. Consequently, spreading-factor selection requires a balance between communication range and transmission speed. The actual distance achieved depends on the operating environment, antenna arrangement, interference, and other radio settings.

In this project, Spreading Factors 8 and 9 are considered in the experimental evaluation. Communication distance, bitrate, and transmission time are measured to compare the two

configurations under the conditions used during testing. These measurements provide practical information about the selected radio settings and complement the network-discovery implementation

2.5 Research Gap and Project Motivation

The reviewed literature establishes that LoRa supports long-range communication and that multi-hop networking can extend connectivity beyond direct radio links. Existing research has investigated peer-to-peer mesh networks, relay-based communication, distance-vector routing, and neighbour-based route selection. Studies have also examined how spreading factor and other physical-layer settings affect communication performance.

However, routing approaches differ in their control structure and implementation requirements. Distributed routing protocols maintain route information across nodes, while the approach considered in this project begins with a master-controlled discovery process and delegates additional searches to selected responding nodes when direct discovery is incomplete.

The project therefore focuses on implementing and examining this cooperative discovery approach using four ESP32 microcontrollers and Reyax RYLR998 LoRa modules. Its experimental component compares Spreading Factors 8 and 9 using measured communication distance, bitrate, and transmission time.

The main project contribution is the practical investigation of a master-coordinated discovery and delegated-search mechanism alongside the measurement of LoRa communication parameters on the selected hardware. The prototype is small-scale, and its findings should be interpreted within the conditions of the experiments conducted rather than treated as proof of performance in large mobile networks or operational military environments.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 Existing Communication Systems

Wireless communication between mobile vehicles is commonly achieved through cellular networks, Wi-Fi, satellite communication, or direct radio links. These technologies provide different levels of communication range, data rate, and infrastructure dependency.

Cellular and Wi-Fi networks generally depend on supporting infrastructure, which may not be available in remote or isolated environments. Direct wireless communication allows two devices to exchange information without an access point, but communication becomes difficult when the destination is outside the sender's direct radio range.

LoRa technology offers an alternative for exchanging small amounts of information over long distances with relatively low power consumption. However, establishing LoRa communication alone does not provide the specific node-discovery and cooperative-search mechanism required by this project. These functions must be handled by the communication protocol implemented on the microcontrollers.

The present project addresses this limitation by developing a master-coordinated discovery mechanism using ESP32 microcontrollers and Reyax RYLR998 LoRa modules.

3.2 Drawbacks of Existing Systems

The main limitations of conventional and basic direct-communication approaches are:

1. **Infrastructure dependency:** Communication through cellular networks or Wi-Fi access points may be unavailable in remote areas.

2. **Limited direct connectivity:** A node outside the direct communication range of another node may not receive its messages.

3. **Limited discovery capability:** A simple direct-link system may not include a mechanism for searching for nodes that fail to respond to the master.

4. **Lack of cooperative searching:** Without an appropriate protocol, reachable nodes cannot automatically assist the master in discovering other nodes.

5. **Radio parameter trade-offs:** LoRa communication performance varies with spreading factor and other radio settings, affecting data rate and transmission time.

These limitations motivate the development of a multi-node communication prototype that combines LoRa radio links with a master-coordinated discovery procedure.

3.3 Proposed Dynamic LoRa Mesh Network

The proposed system is a four-node LoRa communication prototype consisting of one master node and three additional nodes. Each node uses an ESP32 microcontroller interfaced with a Reyax RYLR998 LoRa module.

The ESP32 executes the application-level communication protocol, while the RYLR998 module handles wireless transmission and reception. The modules are configured through AT commands sent over UART. The firmware is developed using the Espressif IoT Development Framework (ESP-IDF).

The master initiates network discovery by transmitting a discovery message. Nodes that receive the message respond with acknowledgements, allowing the master to identify the nodes that have responded directly. If some expected nodes remain undiscovered, the master can send search requests to selected responding nodes. These nodes act as submasters and assist in searching for the missing nodes.

The proposed system operates through the following steps:

1. **Node initialization:** The ESP32 initializes the UART interface and configures the associated LoRa module.

2. **Network discovery:** The master broadcasts a discovery message to identify reachable nodes.

3. **Acknowledgement collection:** Available nodes respond to the master, and their identifiers are recorded.

4. **Identification of missing responses:** The master maintains a list of nodes that have not responded within the discovery period.

5. **Cooperative search:** The master delegates a search request to selected responding nodes.

6. **Search response:** The selected nodes attempt to discover the remaining nodes and report their findings according to the implemented protocol.

7. **Network update:** The master updates its record of discovered nodes based on the received responses.

8. **Retransmission, Timeout and Randomized Backoff:** The protocol uses a maximum of five discovery attempts and a response timeout of 3000 ms . Slave nodes use a randomized backoff delay between 100 ms and 1000 ms before transmitting responses. These mechanisms help manage missed responses, limit repeated attempts, and reduce collisions between simultaneous transmissions.

This mechanism aims to extend node discovery beyond the master's direct communication links. Establishing usable multi-hop data routes additionally requires packet-forwarding logic between intermediate nodes.

3.4 Advantages of the Proposed System

The proposed system offers the following potential advantages:

1. **Reduced infrastructure dependency:** The nodes communicate through LoRa radio links without requiring a cellular network or Wi-Fi access point.

2. **Cooperative node discovery:** Reachable nodes can assist the master in searching for nodes that do not respond directly.

3. **Flexible network expansion:** The design can be extended to additional nodes by modifying the addressing and discovery procedures.

4. **Configurable radio operation:** LoRa parameters can be adjusted to examine different communication conditions.

5. **Embedded implementation:** The ESP32 and Reyax RYLR998 provide a practical platform for implementing and testing the communication protocol.

6. **Experimental evaluation:** The prototype allows communication distance, bitrate, and transmission time to be compared at Spreading Factors 8 and 9.

3.5 System Block Diagram

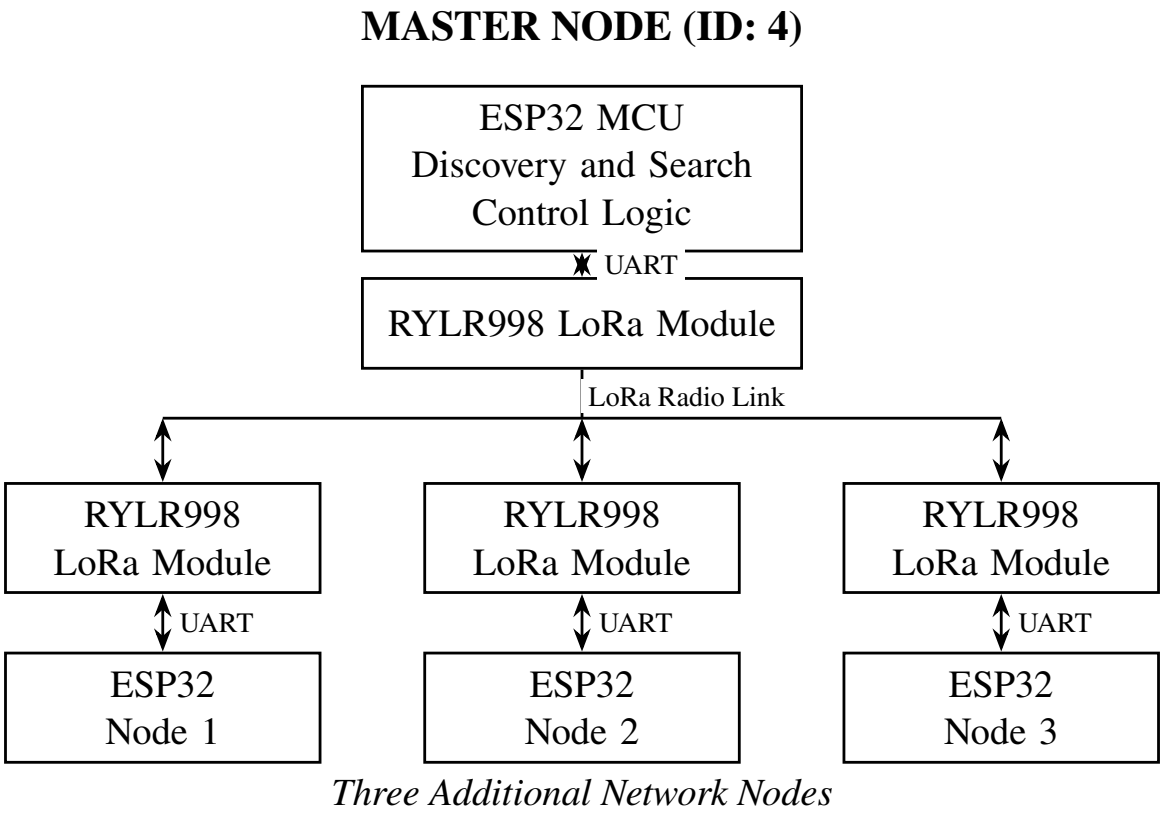


Figure 3.1: Block diagram of the proposed LoRa communication system.

In each node, the ESP32 processes received messages and determines the appropriate response. The RYLR998 module transmits and receives these messages over the wireless link. The master manages discovery and search requests, while the other nodes respond to discovery messages and can assist with the cooperative search procedure.

The slave nodes use a randomized backoff delay before transmitting responses to help reduce collisions between simultaneous transmissions. The master collects acknowledgements and updates its record of discovered nodes based on the responses received. Communication between the ESP32 and the RYLR998 module is handled through the UART interface.

The diagram represents the logical network architecture. Actual communication links depend on radio range and the implemented protocol.

3.6 Network Discovery and Routing Flowchart

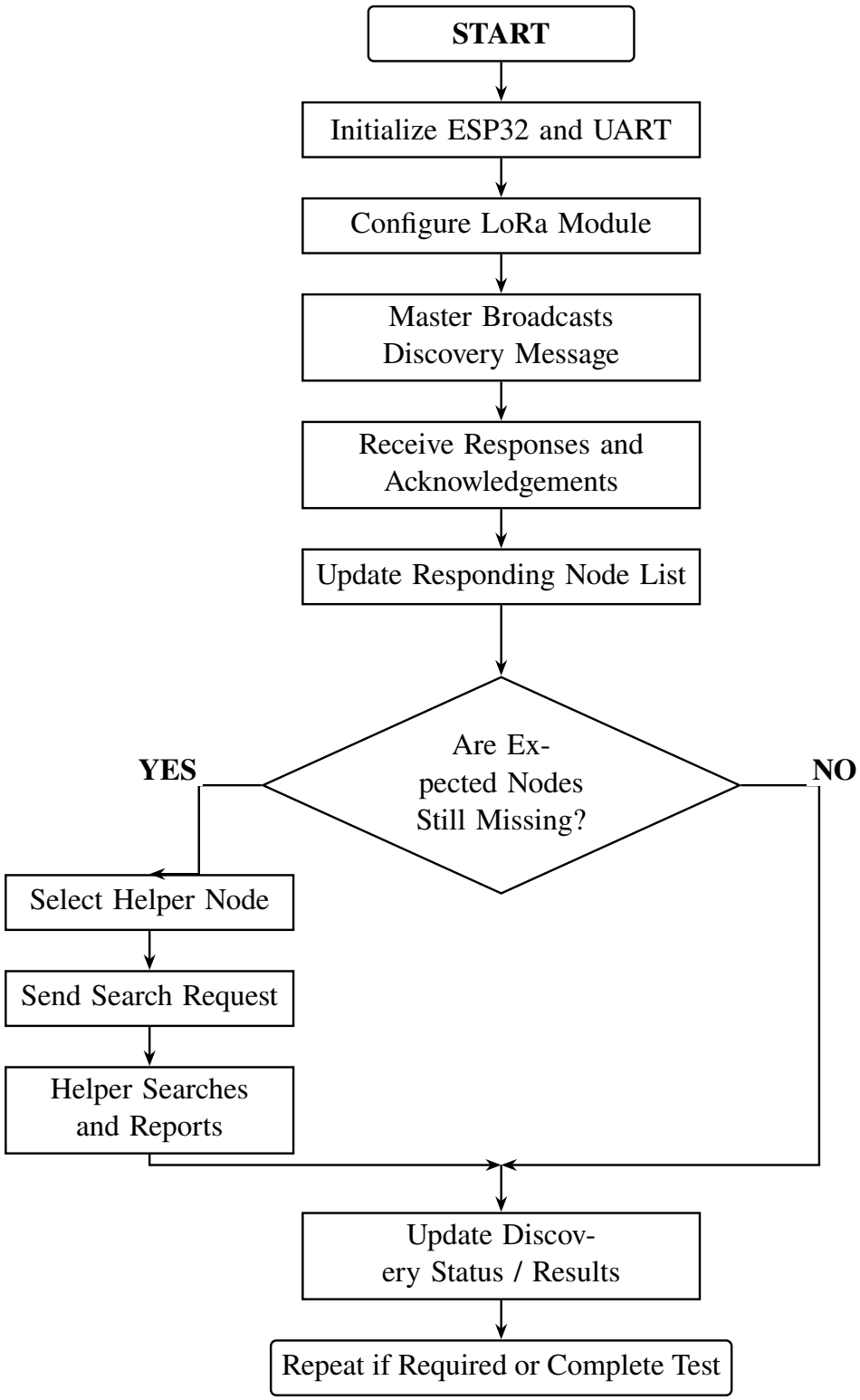


Figure 3.2: Flowchart of the master-coordinated discovery procedure.

3.7 Hardware Requirements

ESP32-C6 Development Board

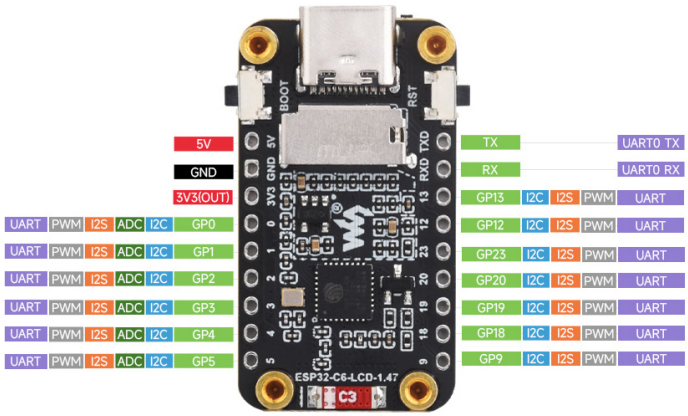


Figure 3.3: ESP32-C6 development board and pin configuration.

The ESP32-C6 is a low-power system-on-chip (SoC) developed by Espressif Systems. It supports Wi-Fi 6 connectivity and uses a RISC-V processor architecture, making it suitable for embedded and IoT applications.

RYRL998 LORA MODULE



Figure 3.4: ryrl998 module with antenna

Four RYLR998 LoRa modules provide wireless communication between the nodes. They support long-range data transmission and are controlled by the ESP32 through UART-based AT commands.

USB CABLE



Figure 3.5: usbcable.

USB cables are used for powering, programming, and debugging the ESP32 boards.

3.8 Software Requirements

1. **Visual Studio Code:** Visual Studio Code is used as the integrated development environment for writing, editing, building, and managing the ESP32 firmware.

2. **ESP-IDF:** The Espressif IoT Development Framework (ESP-IDF) provides the tools and libraries required to develop, compile, and flash firmware onto the ESP32 boards.

3. **UART Communication Interface:** UART enables serial communication between the ESP32 and the RYLR998 module. It is used to send AT commands and receive module responses and incoming communication messages.

4. **RYLR998 AT Command Interface:** AT commands are used to configure module parameters such as device address, network ID, frequency, and radio settings. These commands allow the LoRa modules to be configured for communication within the four-node network.

5. **Serial Monitor:** The serial monitor is used to observe module responses, display debugging messages, and verify communication during node discovery and experimental testing.

CHAPTER 4

METHODOLOGY

4.1 Hardware Setup and LoRa Module Configuration

CIRCUIT

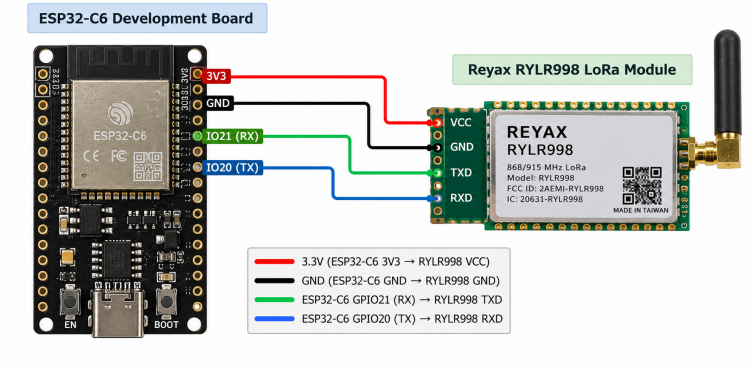


Figure 4.1: Esp32-c6 with ryrl998 module.

4.1.1 LoRa Radio Parameter Configuration

The RYLR998 LoRa modules were configured using the `AT+PARAMETER` command. The command accepts four parameters: spreading factor, bandwidth setting, coding rate, and programmed preamble length.

Table 4.1: Detailed RYLR998 radio parameter configuration

Parameter	**Value**	**Description**
Spreading Factor	8	SF8 is used for LoRa transmission.
Bandwidth Setting	7	Corresponds to a bandwidth of 125 kHz.
Coding Rate	1	Represents a coding rate of 4/5.
Preamble Length	12	The programmed preamble length is 12 symbols.

The radio parameters were configured using the following AT command:

```
AT+PARAMETER=8,7,1,12
```

The bandwidth setting of 125 kHz and coding rate of 4/5 were used along with spreading factor 8 and a preamble length of 12 symbols. These settings define the radio configuration used during the corresponding experiments.

Table 4.2: Overall RYLR998 module configuration

Configuration	**Value**
Network ID	18
Operating Frequency	915 MHz
Spreading Factor	8
Bandwidth	125 kHz
Coding Rate	4/5
Preamble Length	12 symbols
UART Baud Rate	115200 bps
Master Address	4
Other Node Addresses	1, 2, 3

4.2 ESP32 Firmware Development Using ESP-IDF

The firmware is developed in C using the Espressif IoT Development Framework (ESP-IDF) in Visual Studio Code. The ESP32 executes the application-level communication protocol, while the RYLR998 module handles radio transmission and reception.

The firmware initializes the UART interface, sends AT commands, receives module responses, and processes incoming messages. Separate program logic is used for the master and slave nodes according to their assigned roles.

The master manages the discovery process and search requests. Slave nodes process incoming messages, send acknowledgements, and perform the response procedures defined by the protocol. The firmware also incorporates response timeouts, a limited number of discovery attempts, and randomized backoff at slave nodes.

4.3 Packet Structure and Communication Protocol

The proposed LoRa network uses a set of control packets to manage node discovery, acknowledgements, cooperative searching, and forwarding operations. These control packets coordinate communication between the master and the other network nodes. After the required nodes have been discovered, data packets can be transmitted to the intended destination.

The ESP32 microcontrollers implement the application-level communication protocol, while the Reyax RYLR998 modules handle wireless transmission and reception.

4.3.1 Control Packet Types

The control packets used in the node discovery and communication process are listed in Table 4.3.

Table 4.3: Control packet types and their functions

Packet Type	**Purpose**
`DISCOVERY`	Broadcasts a discovery message to identify reachable nodes.
`ACK_DISCOVERY`	Acknowledges the receipt of a discovery message.
`SEARCH`	Requests a responding node to search for nodes that have not yet been discovered.
`ACK_SEARCH`	Acknowledges the receipt of a search request.
`PROBE`	Supports the probing process used to search for target nodes.
`ACK_PROBE`	Provides a response to a probe message.
`RESULT:F:N`	Reports the search result using the result fields defined in the implemented protocol.
`FWD`	Carries a message that requires forwarding according to the implemented protocol.
`ACK_FWD`	Acknowledges a forwarding message.
`FWD_FAIL`	Indicates that a forwarding operation has failed.

4.3.2 Node Discovery and Cooperative Search

The master node initiates the discovery process by broadcasting a `DISCOVERY` packet. Nodes that receive the message respond with `ACK_DISCOVERY` packets. The master records the responding nodes and identifies nodes that have not responded within the discovery period.

When some expected nodes remain undiscovered, the master can send a `SEARCH` request to selected responding nodes. These nodes assist in the search process using the `PROBE` and `ACK_PROBE` packets. The resulting information is reported using the result message defined by the protocol.

The master uses the received responses to update its record of discovered nodes. The control packets therefore support the coordination of node discovery and cooperative searching within the network.

4.3.3 Data Packet Structure

After node discovery, the network can transmit application data to an intended destination. The data packet contains the destination address, the length of the data, and the data payload.

The basic logical structure of the data packet is shown in Table 4.4.

Table 4.4: Logical structure of a data packet

Field	**Description**	**Function**
1	Destination Address	Identifies the node to which the data packet is addressed.
2	Data Length	Specifies the length of the data payload.
3	Data Payload	Contains the application data to be transmitted to the destination node.

The logical data packet format can be represented as follows:

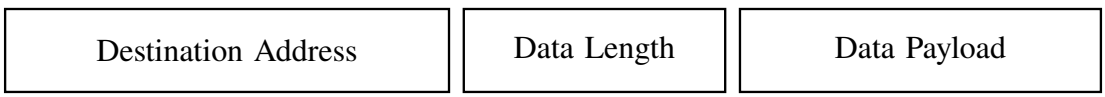


Figure 4.2: Logical structure of the sender data packet.

Payload Length: The application-level payloads used in the proposed communication protocol typically range from 5 to 13 bytes, depending on the packet type and the information

being transmitted. The payload contains control messages, node identifiers, search requests, acknowledgements, or forwarding information. Keeping messages compact helps reduce the amount of data transmitted over the LoRa link.

The destination address identifies the intended receiver, while the data length indicates the amount of payload data being transmitted. The payload contains the actual application information. The receiver uses the destination address and length information to identify the intended destination and determine the payload size.

4.3.4 Received Data Packet Structure

The receiver processes incoming data packets containing the sender address, data length, data payload, and sequence number. The sender address identifies the node that transmitted the packet, while the data length specifies the size of the payload. The sequence number helps distinguish individual packets and can be used to identify repeated packets or track packet order.

Table 4.5: Structure of the received data packet

No.	Field	Description
1	Sender Address	Identifies the node that transmitted the packet.
2	Data Length	Specifies the length of the data payload.
3	Data Payload	Contains the application data transmitted by the sender.

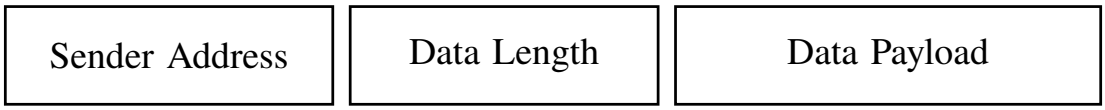


Figure 4.3: Logical structure of the received data packet.

4.3.5 Communication Procedure

The overall communication procedure consists of two main stages:

1. **Node discovery stage:** The master broadcasts discovery packets, collects acknowledgements, and coordinates the cooperative search for nodes that have not responded.

2. **Data transmission stage:** After the required nodes have been discovered, data packets are addressed to the intended destination using the destination address, data length, and payload fields.

The control packet format and the data packet format serve different purposes. Control packets manage network discovery and protocol coordination, whereas data packets carry the application information intended for a destination node.

4.4 Master-Based Node Discovery and Acknowledgement Handling

The discovery process begins when the master broadcasts a discovery message. Slave nodes that receive this message prepare their responses according to the implemented protocol.

To reduce simultaneous transmissions, the slave nodes use a randomized backoff interval between 100 ms and 1000 ms before sending applicable responses. This spreads response transmissions over different time intervals and reduces the likelihood of collisions between neighbouring nodes.

The master collects acknowledgements within a discovery response window of 3000 ms. Nodes that do not respond within this period are marked as unresponsive for that discovery attempt. The discovery process is configured for a maximum of five attempts using MAXTRIES=5.

After collecting the responses, the master updates its record of responding and undiscovered nodes. This information is used to determine whether further discovery attempts or cooperative searching are required.

4.5 Cooperative Search and Dynamic Route Recovery

When some expected nodes remain undiscovered after the initial discovery process, the master identifies the unresolved node IDs and prepares a search request. The request is sent to selected responding nodes, which act as submasters and assist in the search procedure.

The selected nodes execute the search operations specified in the firmware and report the results to the master. Based on the received information, the master updates its record of discovered nodes.

This approach allows reachable slave nodes to assist the master instead of relying exclusively on direct discovery. It forms the basis of the proposed cooperative discovery

mechanism. Any claim of complete dynamic route recovery or multi-hop data forwarding must correspond to the forwarding and routing functions actually implemented in the firmware.

4.6 Experimental Setup for Spreading Factors 8 and 9

The experimental evaluation examines the communication performance of the Reyax RYLR998 modules using Spreading Factors 8 and 9.

The selected spreading factor is configured through the module's AT command interface. For a fair comparison, the other radio settings and test conditions should be kept consistent wherever possible. Wireless communication is then tested at the selected distances, and the observations are recorded for each configuration.

The experiments focus on the following parameters:

Communication distance.

Bitrate.

Time-on-Air (ToA) or transmission time, as recorded during the experiments. The observations for Spreading Factors 8 and 9 are compared to understand the effects of the selected spreading factor on communication performance in the test environment.

4.7 Measurement Procedure for Distance, Bitrate and Transmission Time

The experimental evaluation was carried out using four ESP32-C6 boards interfaced with Reyax RYLR998 LoRa modules. A scooter was used as a mobile platform to vary the distance between the communicating nodes and examine the network's communication capability under different conditions. The experiments were conducted using Spreading Factors 8 and 9.

Three test scenarios were considered to examine direct communication and cooperative node discovery.

4.7.1 Case 1: Direct Communication Test

In the first experiment, the master node attempted to communicate directly with all three slave nodes. The scooter was used to increase the separation distance and observe whether

the master could receive acknowledgements from every slave node.

The maximum distance at which acknowledgements were successfully received from all three slave nodes was approximately 450 metres for SF8 and 480 metres for SF9. These observations were used to establish the direct communication range under the tested conditions.

4.7.2 Case 2: Cooperative Search Using One Responding Node

In the second experiment, one slave node was positioned approximately 450 metres from the master, while the other two nodes were outside the master's direct communication range. Only the reachable slave node responded to the master's discovery message.

The master then sent a search request to the responding node, which acted as a relay node and attempted to discover the remaining two nodes. The master waited for 15 seconds to receive the search results. If no result was received within this period, another search attempt was initiated, with a maximum of two additional retries.

4.7.3 Case 3: Cooperative Search Using Two Responding Nodes

In the third experiment, two slave nodes responded directly to the master's discovery message, while the remaining node did not respond directly. The master used the two responding nodes to assist in searching for the undiscovered node through the cooperative search procedure.

The responses from these nodes were observed to examine how multiple reachable nodes could participate in the discovery process and assist the master in locating an unresponsive node.

4.7.4 Calculation of Bitrate and Transmission Time

For each spreading factor, the bitrate and packet transmission time were calculated using the LoRa communication equations presented below.

Bitrate Calculation

The bitrate is calculated based on the spreading factor (SF), bandwidth (BW), and coding rate (CR):

$$R_b = \frac{SF \times BW}{2^{SF} \times CR} \tag{4.1}$$

where:

- R_b represents the bitrate.
- SF represents the spreading factor.
- BW represents the bandwidth.
- CR represents the coding rate.

Time-on-Air Calculation

The packet transmission time, or Time-on-Air (ToA), is estimated using the payload length, preamble length, and calculated bitrate:

$$T_{\text{airtime}} = \left(n_{\text{payload}} + n_{\text{preamble}} + 4.25\right) \times \frac{1}{R_b} \tag{4.2}$$

where:

- T_{airtime} represents the packet transmission time.
- n_{payload} represents the payload length.
- n_{preamble} represents the preamble length.
- R_b represents the calculated bitrate.

The calculated values were used to compare the communication characteristics of SF8 and SF9. The experimental observations from all three test cases are presented in Chapter 5.

CHAPTER 5

TEST CASES AND RESULTS

5.1 Test Case Design and Experimental Procedure

Three test cases were considered to evaluate direct communication and cooperative node discovery using four ESP32-C6 boards and Reyax RYLR998 LoRa modules. The experiments were conducted using Spreading Factors 8 and 9.

Test Case 1: Direct Communication

Objective: To determine the maximum distance at which the master can receive acknowledgements directly from all three slave nodes.

Initial Conditions: One master and three slave nodes were configured with their respective addresses and identical LoRa settings.

Expected Behaviour: All three slave nodes should acknowledge the master's discovery message within the direct communication range.

Observed Behaviour: Successful acknowledgements from all three slave nodes were received up to approximately 450 metres for SF8 and 480 metres for SF9.

Test Case 2: Cooperative Search Using One Responding Node

Objective: To examine whether one reachable slave node can assist the master in discovering two nodes outside its direct communication range.

Initial Conditions: One slave node was positioned approximately 450 metres from the master, while the other two nodes were placed beyond its direct communication range.

Expected Behaviour: The responding node should act as a submaster, search for the two undiscovered nodes, and report the results to the master.

Observed Behaviour: The master received an acknowledgement from one slave node and initiated the cooperative search process. The responding node attempted to discover the remaining two nodes.

Test Case 3: Cooperative Search Using Two Responding Nodes

Objective: To examine how two reachable slave nodes can assist the master in searching for an undiscovered node.

Initial Conditions: Two slave nodes were placed within the master's direct communication range, while the third node was outside its range.

Expected Behaviour: The master should initially assign the search task to one responding node. If it does not receive an acknowledgement within one second after the permitted two retries, it should select the next responding node.

Observed Behaviour: Both slave nodes responded directly to the master. The master initiated the search procedure and followed the configured acknowledgement timeout, retransmission, and node-selection mechanism.

5.2 Node Discovery and Communication Test Cases

Table 5.1: Comparison of ToA and Bitrate for SF8 and SF9

Parameter	**SF8**	**SF9**
Minimum ToA (5 bytes)	6.80 ms	12.09 ms
Average ToA (5–13 bytes)	8.08 ms	14.36 ms
Maximum ToA (13 bytes)	9.36 ms	16.64 ms
Bitrate	3125 bps	1757.81 bps

Table 5.1 presents the comparison of Time-on-Air (ToA) and bitrate for Spreading Factors 8 and 9. SF8 provides a higher bitrate and lower transmission time compared to SF9. As the spreading factor increases, the bitrate decreases and the packet transmission time increases.

5.2.1 Experimental Results for Spreading Factor 8

The experimental evaluation was conducted using a spreading factor of 8 (SF8). The LoRa module was configured with a bandwidth setting of 125 kHz, coding rate of 4/5, and a preamble length of 12 symbols.

Table 5.2: Experimental Distance Observations for SF8

Case	Experimental Scenario	Distance Observations
1	**Direct Communication**	Successful communication with all three slave nodes was achieved at an approximate maximum distance of 450 m from the master node.
2	**Cooperative Search Using One Responding Node**	Responding node to master: approximately 450 m. First unresponsive node to responding node: approximately 250 m. Second unresponsive node to responding node: approximately 280 m. Both unresponsive nodes were outside the master's direct communication range, with no direct line of sight between the responding node and the unresponsive nodes.
3	**Cooperative Search Using Two Responding Nodes**	Two responding nodes were approximately 450–460 m from the master node. The remaining unresponsive node was approximately 250–300 m from one of the responding nodes. The cooperative search process helped the master identify all three slave nodes.

5.2.2 Experimental Results for Spreading Factor 9

The experimental evaluation was conducted using a spreading factor of 9 (SF9). The LoRa modules were configured with a bandwidth of 125 kHz, coding rate of 4/5, and a preamble length of 12 symbols. The performance was evaluated in terms of time on air (ToA), bitrate, direct communication distance, and cooperative node discovery.

Table 5.3: Experimental Distance Observations for SF9

Case	Experimental Scenario	Distance Observations
1	**Direct Communication**	Successful communication with all three slave nodes was achieved at an approximate maximum distance of 470 m from the master node.
2	**Cooperative Search Using One Responding Node**	Responding node to master: approximately 450 m. First unresponsive node to responding node: approximately 270 m. Second unresponsive node to responding node: approximately 283 m. Both unresponsive nodes were outside the master's direct communication range, with no direct line of sight between the responding node and the unresponsive nodes.
3	**Cooperative Search Using Two Responding Nodes**	Two responding nodes were approximately 450–480 m from the master node. The remaining unresponsive node was approximately 250–320 m from one of the responding nodes. The cooperative search process helped the master identify all three slave nodes.

5.3 Results and Interpretation

The experiments were conducted using four ESP32-C6 nodes and Reyax RYLR998 LoRa modules with Spreading Factors 8 and 9. Direct communication and cooperative node discovery were tested under different distances and LOS conditions.

5.3.1 Direct Communication

The maximum observed distance for successful communication between the master and all three slave nodes was approximately 450 m for SF8 and 470 m for SF9. SF9 achieved a

slightly greater distance in the conducted experiments.

5.3.2 Cooperative Node Discovery

In Case 2, one responding node assisted the master in searching for two undiscovered nodes located approximately 250 m and 280 m away without direct LOS.

In Case 3, two slave nodes responded directly to the master, and one of them assisted in discovering the remaining node located approximately 250–300 m away. Through this approach, all three slave nodes were discovered.

5.3.3 Effect of Line-of-Sight

LOS conditions have a significant influence on LoRa communication. Clear LOS generally provides better signal reception and more reliable packet delivery, whereas obstacles can weaken signals and result in missed acknowledgements. Under favourable LOS conditions, communication distances may approach 1 km, while NLOS conditions can reduce the achievable range.

5.3.4 Overall Interpretation

The experiments demonstrate that the proposed cooperative discovery mechanism can help the master identify nodes that are outside its direct communication range. SF9 achieved a greater observed distance than SF8 in the direct communication test, although it generally involves a lower bitrate and longer transmission time. The results highlight the importance of spreading factor, node position, and LOS conditions in LoRa communication.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presents a Dynamic LoRa Mesh Network for Military Vehicle Communication using four ESP32-C6 microcontrollers and Reyax RYLR998 LoRa modules. The system implements a master-coordinated communication protocol that includes node discovery, acknowledgement handling, cooperative searching, and forwarding-related message handling.

The main objective was to allow the master node to discover other nodes even when direct communication was not possible. This was demonstrated through experiments in which one or two reachable slave nodes assisted the master in searching for undiscovered nodes. Randomized backoff was also implemented at the slave nodes to reduce collisions during simultaneous transmissions.

The experimental evaluation compared Spreading Factors 8 and 9 using communication distance, bitrate, and transmission time. The maximum observed direct communication distances were approximately 450 m for SF8 and 470 m for SF9. The experiments also demonstrated the importance of node positioning and Line-of-Sight conditions in wireless communication.

Overall, the project demonstrates the feasibility of using LoRa technology and cooperative node discovery for multi-node communication without depending on conventional network infrastructure. The developed prototype provides a foundation for further development of dynamic routing and communication systems for mobile vehicle networks..

6.2 Limitations of the Proposed System

Although the proposed system demonstrates cooperative LoRa communication and node discovery, it has certain limitations:

1. Limited Network Size: The prototype is implemented and tested using only four ESP32-C6 nodes. Its performance with a larger number of nodes has not been evaluated.

2. Limited Communication Range Testing: The experiments were conducted under

specific test conditions, with maximum observed direct communication distances of approximately 450 m for SF8 and 470 m for SF9. These results may vary with terrain, antenna placement, obstacles, and interference.

3. Environmental Constraints: Communication performance is affected by Line-of-Sight conditions. Obstacles can weaken radio signals and reduce the reliability of communication.

4. Limited Performance Evaluation: The experimental analysis focuses on communication distance, bitrate, and transmission time. Other metrics, such as Packet Delivery Ratio and end-to-end delay, have not been measured.

5. Limited Mobility Testing: The experiments were conducted using a scooter as a mobile platform. Extensive testing with multiple moving vehicles under different operating conditions has not been performed.

6. Security Considerations: Encryption, authentication, and protection against intentional interference have not been implemented or evaluated.

7. Routing Improvements: The current prototype uses a master-coordinated cooperative discovery approach. Further improvements are needed to support more scalable and robust routing in larger networks.

These limitations define the current scope of the prototype and provide opportunities for future development.

6.3 Future Scope

The following improvements can be considered for future development:

1. Network Expansion: Increase the number of nodes to evaluate the scalability of the proposed communication system.

2. Improved Dynamic Routing: Develop better route-selection mechanisms using RSSI, SNR, and node position to improve communication between nodes.

3. Performance Evaluation: Measure Packet Delivery Ratio (PDR), end-to-end delay, and network recovery time to evaluate communication reliability.

4. Real-Time Mobility Testing: Test the network with multiple moving vehicles to study connectivity and route changes under different operating conditions.

5. Security Enhancement: Implement encryption and authentication to protect communication between network nodes.

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